AEROSPACE ENGINEERING INTELLIGENCE · WORKFLOW WALKTHROUGH

REV A → B

AeroNexus AI

Drawings, sensor data, safety reports and supplier documents go in. Evidence-backed decisions, signed off by a named engineer, come out.

Presented by Muskan · Free, open data · Llama on Groq

THE PROBLEM

One drawing change ripples through orders, work orders, suppliers and customers, and someone traces it by hand.

Revisions hide in drawings

A thickness change from 5 mm to 6 mm is one number on one sheet, easy to miss.

Impact lives in other systems

Sales orders, work in progress, material POs and customer accounts all have to move.

Decisions need evidence

Who approved it, on which numbers, under which version of the rule?

AeroNexus turns each change into a **case**: computed with receipts, reviewed by a named approver, replayable from its log.

AT A GLANCE

What the platform delivers

11.6

cycles RUL error, NASA C-MAPSS FD001

0.92

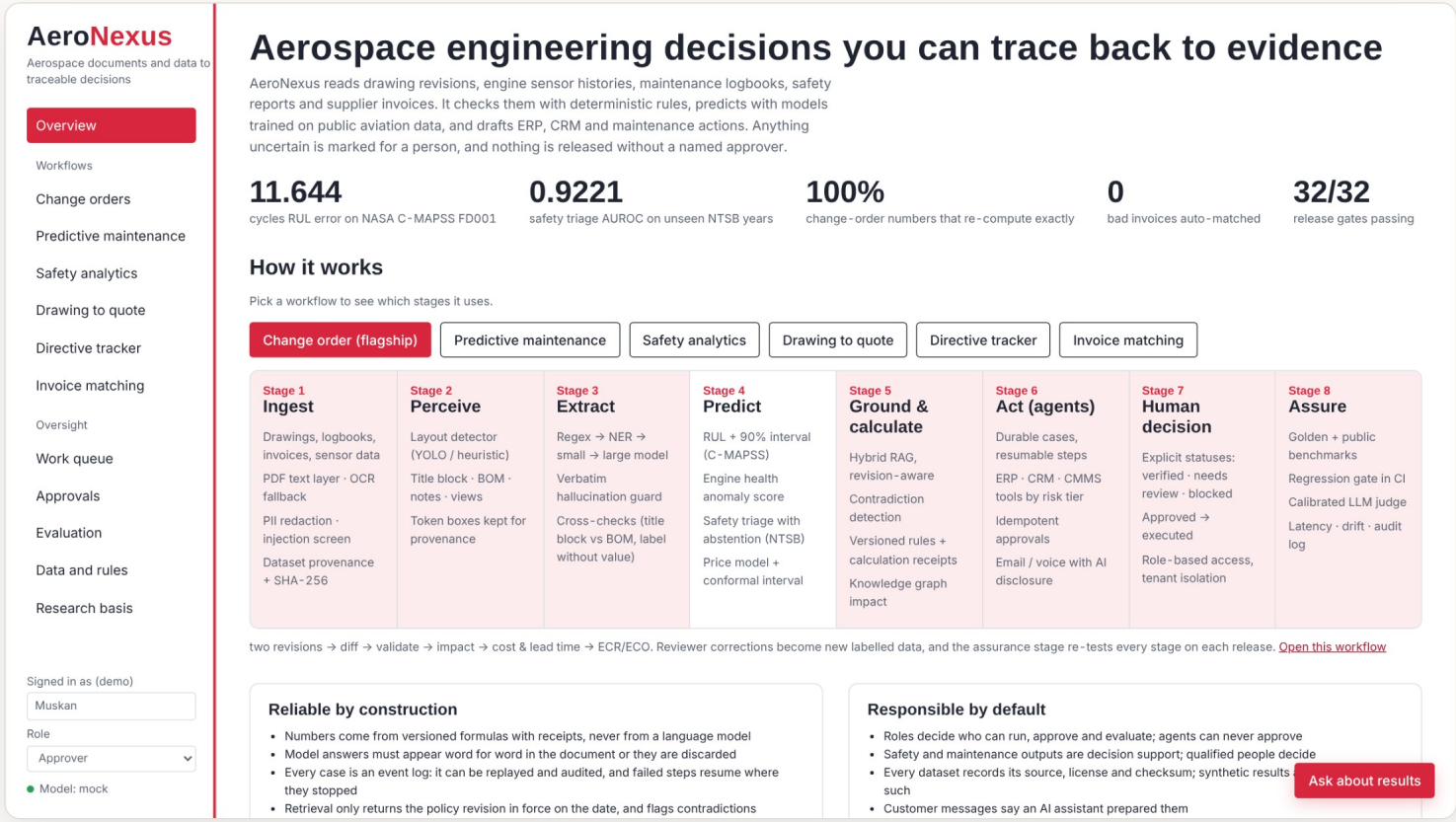
safety triage AUROC, unseen NTSB years

100%

change-order figures re-compute exactly

32/32

release gates passing



ARCHITECTURE

Eight stages, one pipeline

STAGE 1

Ingest

Drawings, logbooks, invoices, sensor data and NTSB text; PII redacted, injections screened.

STAGE 2

Perceive

Layout detector, PDF text layer with OCR fallback, title block and BOM tables.

STAGE 3

Extract

Regex, then NER, then small and large LLMs, with a verbatim hallucination guard.

STAGE 4

Predict

RUL with a 90% conformal range, engine health, safety triage that can abstain.

STAGE 5

Ground and calculate

Hybrid RAG with checked citations; versioned formulas that emit receipts.

STAGE 6

Act

Agents draft ERP, CRM, CMMS and email actions by risk tier, idempotently.

STAGE 7

Human decision

A named approver signs off; agents never can. Roles and tenants enforced.

STAGE 8

Assure

32 release gates, an LLM-judge harness, drift and calibration checks, audit log.

WORKFLOW 1 · ENGINEERING CHANGE ORDERS

Change orders, step by step

01 **Extract** both revisions

02 **Diff** field by field

03 **Validate** revision B

04 **Impact:** affected records

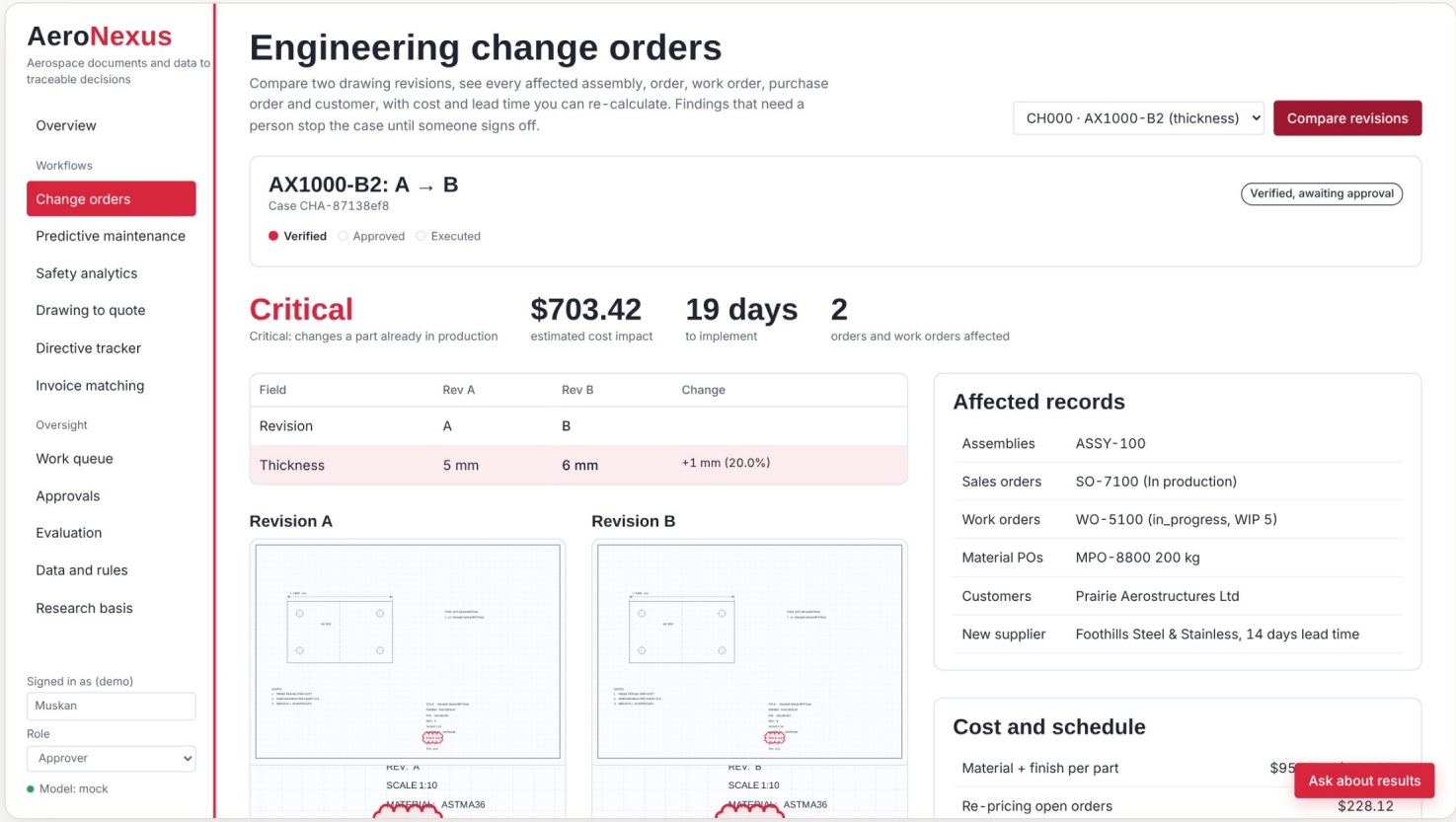
05 **Cost** with formulas

06 **Classify** severity

07 **Approver** signs off

08 **Execute** in ERP, notify

Steps 1–6 run automatically and resume after a failure. Step 7 is always a person.



WORKFLOW 1 · ENGINEERING CHANGE ORDERS

One case: AX1000-B2, 5 → 6 mm

Severity: Critical

The part is already in production on order SO-7100.

Cost: $703.42

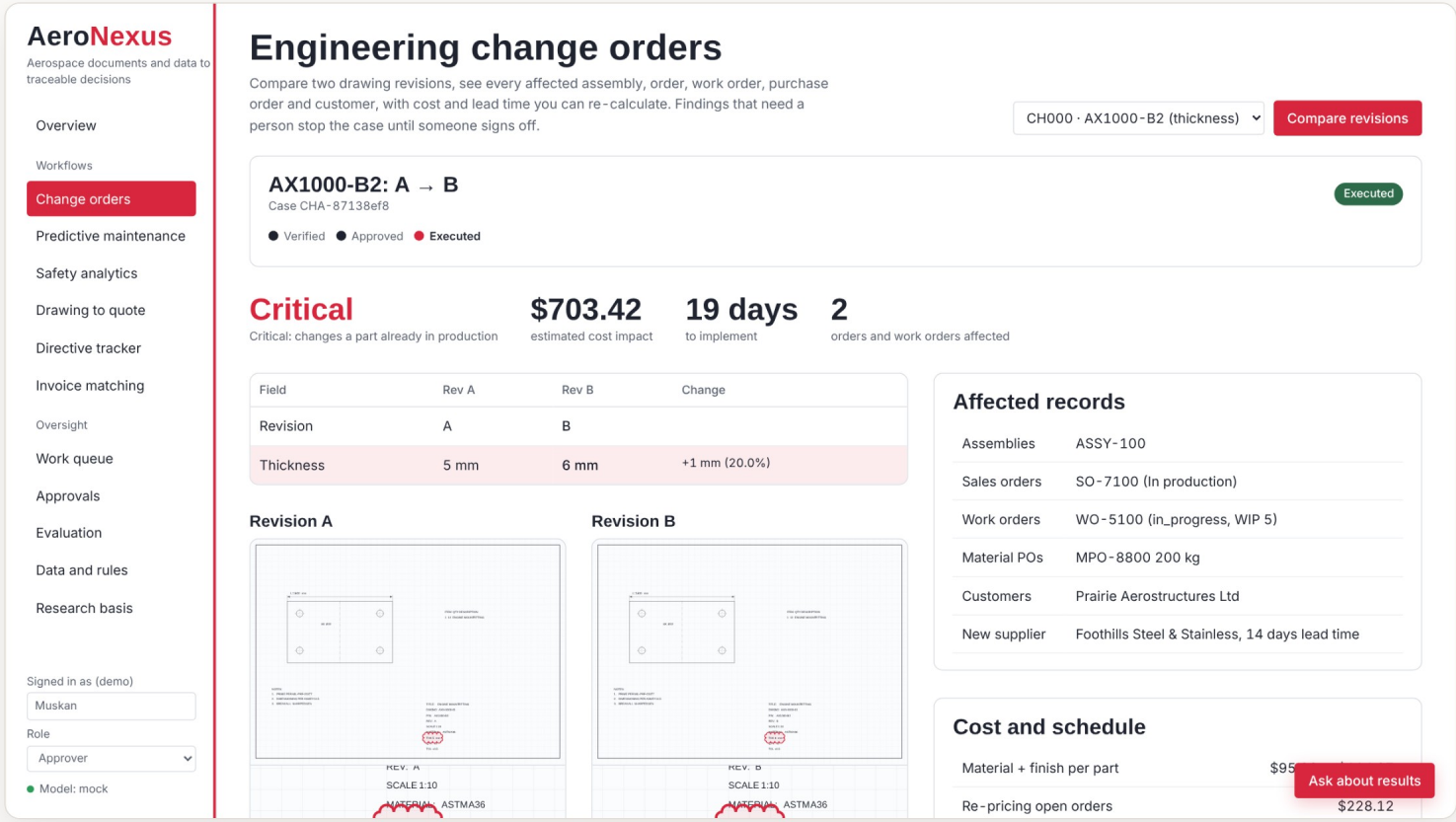
Re-pricing, scrap of work in progress and obsolete stock, net.

Lead time: 19 days

The 6 mm stock needs a new supplier with 14 days' lead.

Affected records

ASSY-100, WO-5100, MPO-8800 and one customer account.



WORKFLOW 1 · ENGINEERING CHANGE ORDERS

Every number has a receipt

No LLM arithmetic

Costs, lead times and severity come from versioned formulas.

Receipts

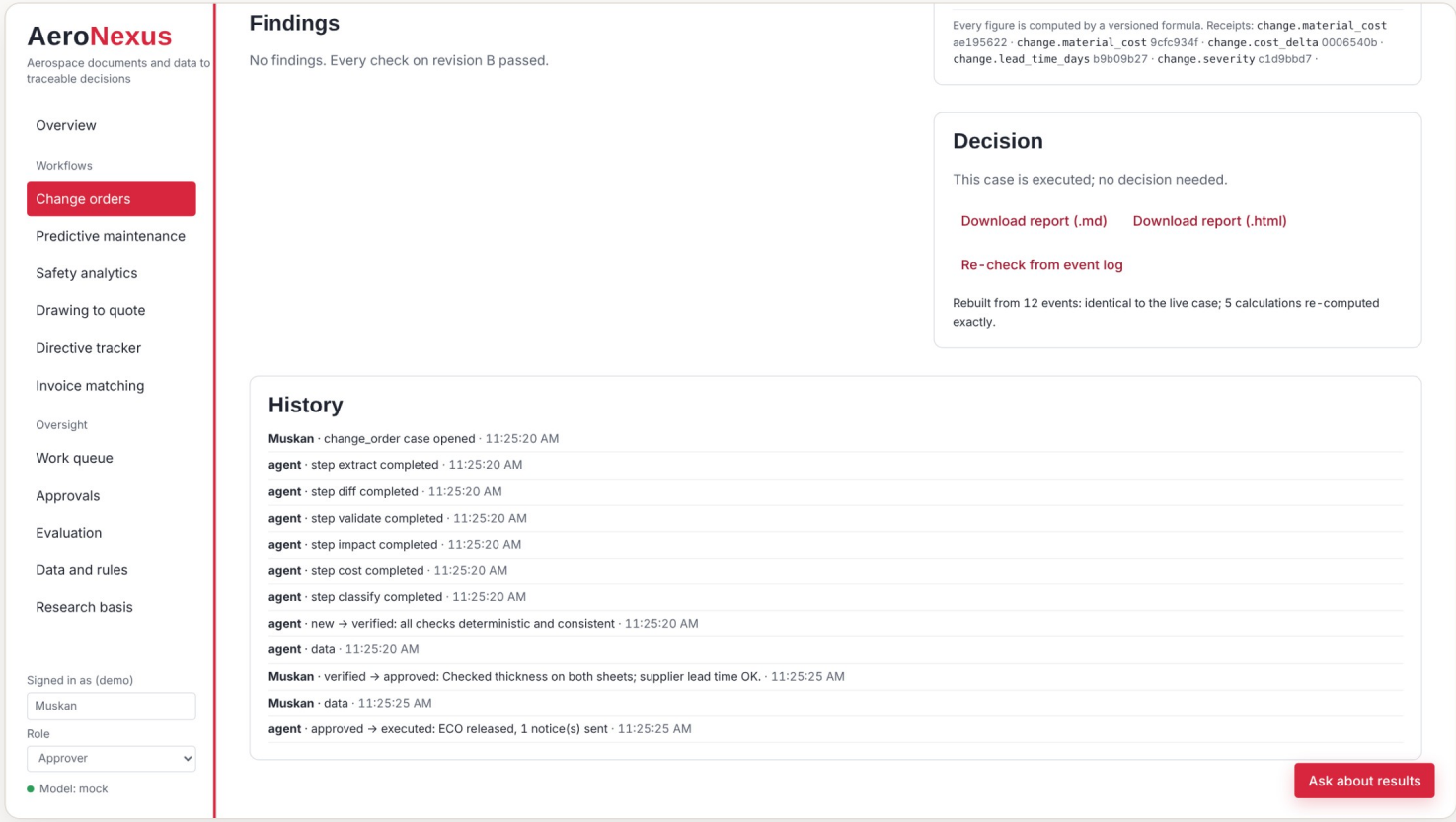
Inputs, rule version and result are hashed and re-run exactly.

Append-only event log

Every step, decision and status change is an event.

Replay check

“Rebuilt from 12 events: identical; 5 calculations re-computed exactly.”

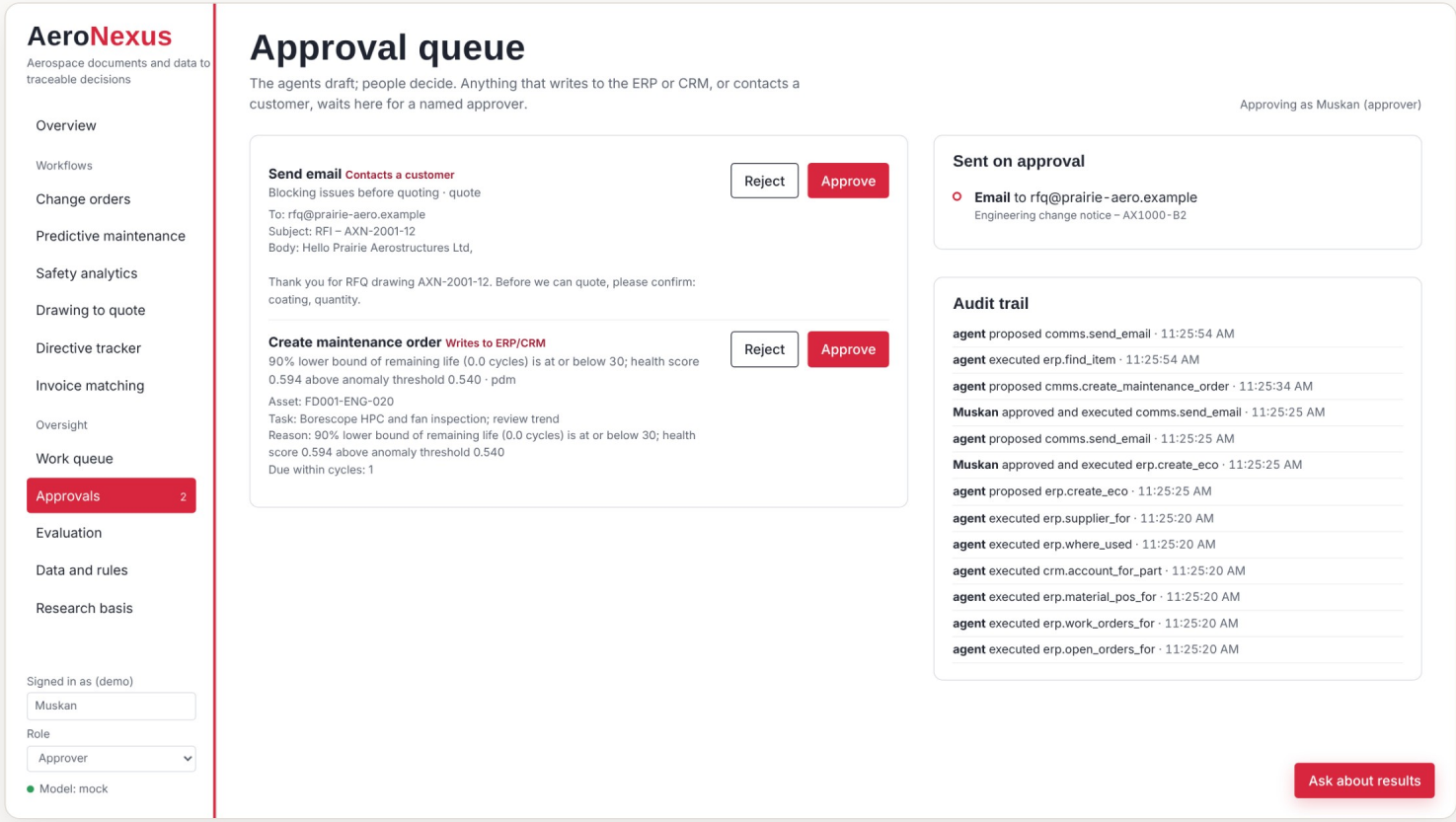


HUMAN IN THE LOOP

People approve; agents only propose

Role	Can
Viewer	Read cases and results
Engineer	Open cases, propose actions
Approver	Approve, reject, release
Admin	Run evaluations, edit rules
Agent	Draft only, never approve

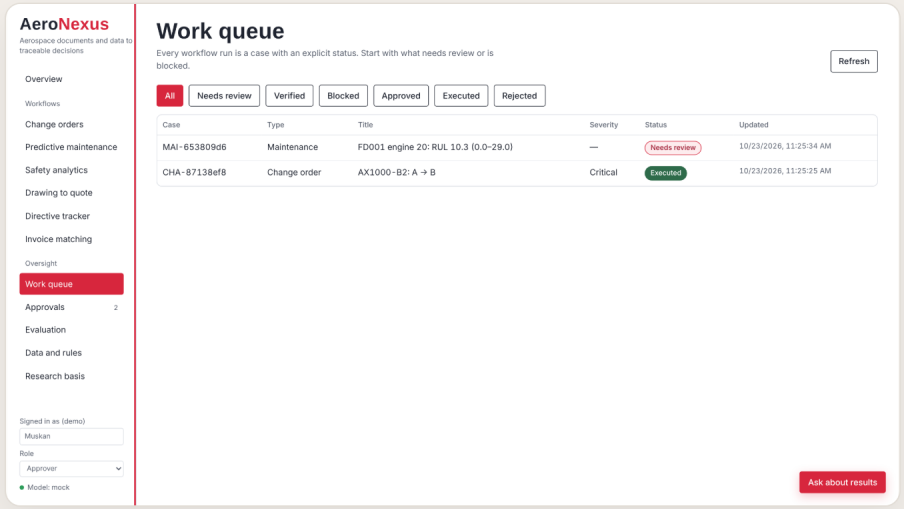
Actions are tiered: read, draft, write, external. Each has an idempotency key, an audit entry and, when outbound, an AI disclosure.



CASE MANAGEMENT

Every run is a case with a status





Allowed moves only

A transition table rejects jumps such as blocked straight to executed.

Retry and resume

Each step is checkpointed; a failed step retries without redoing the rest.

Tenant isolation

A case is visible only inside its own tenant.

WORKFLOW 2 · PREDICTIVE MAINTENANCE

Remaining life, with an honest range

Public data

NASA C-MAPSS turbofan runs, all four subsets, checksum-verified.

Model

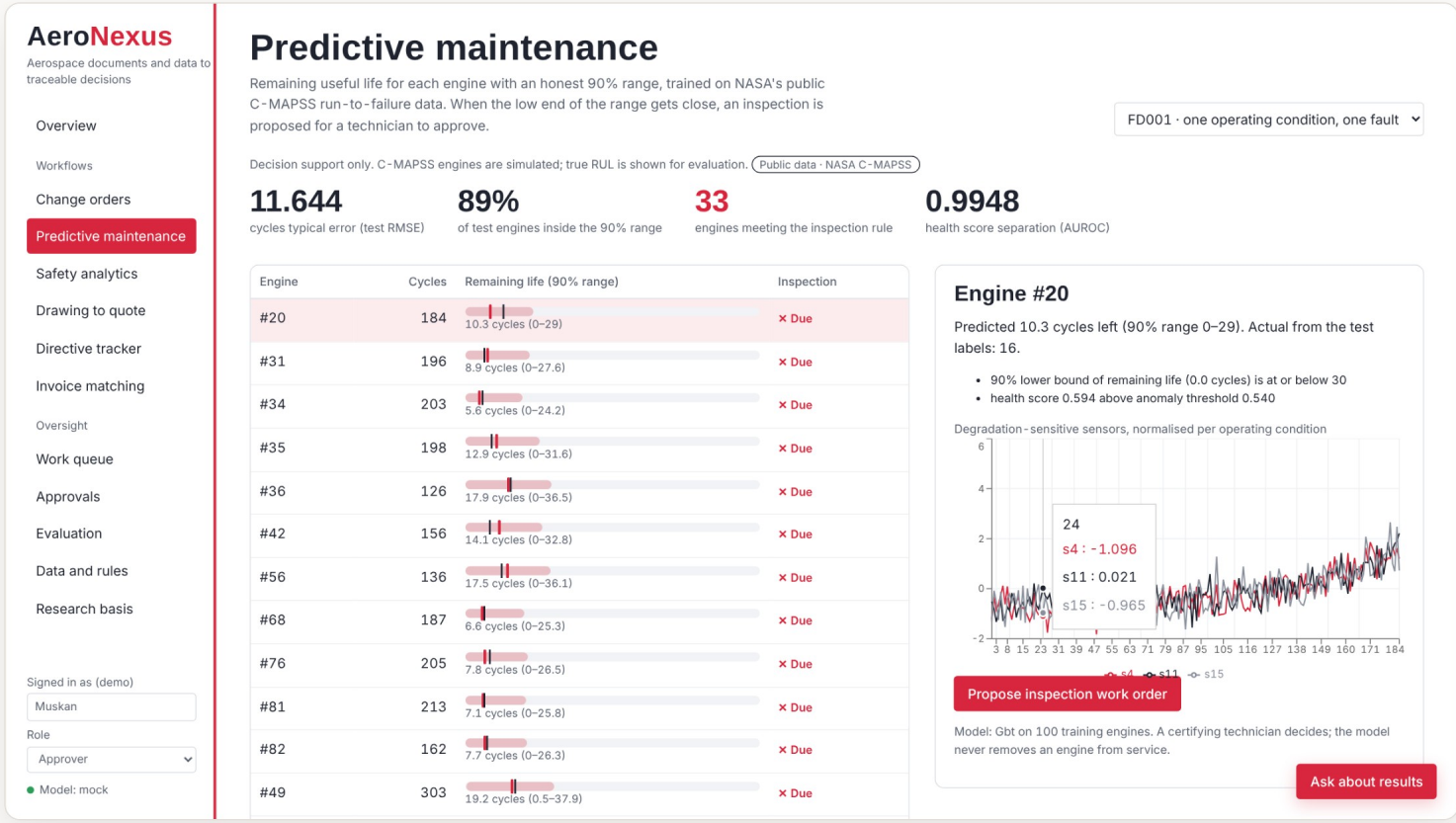
Per-regime normalised windows; gradient-boosted trees beat an MLP.

Range, not a point

Split-conformal 90% interval; test RMSE 11.6–13.5 cycles.

Action

A versioned rule proposes an inspection; a technician decides.



WORKFLOW 3 · SAFETY ANALYTICS

Triage 17,485 NTSB reports

No outcome leakage

Words like “fatal” are masked; AUROC moves by only 0.004.

Split by time

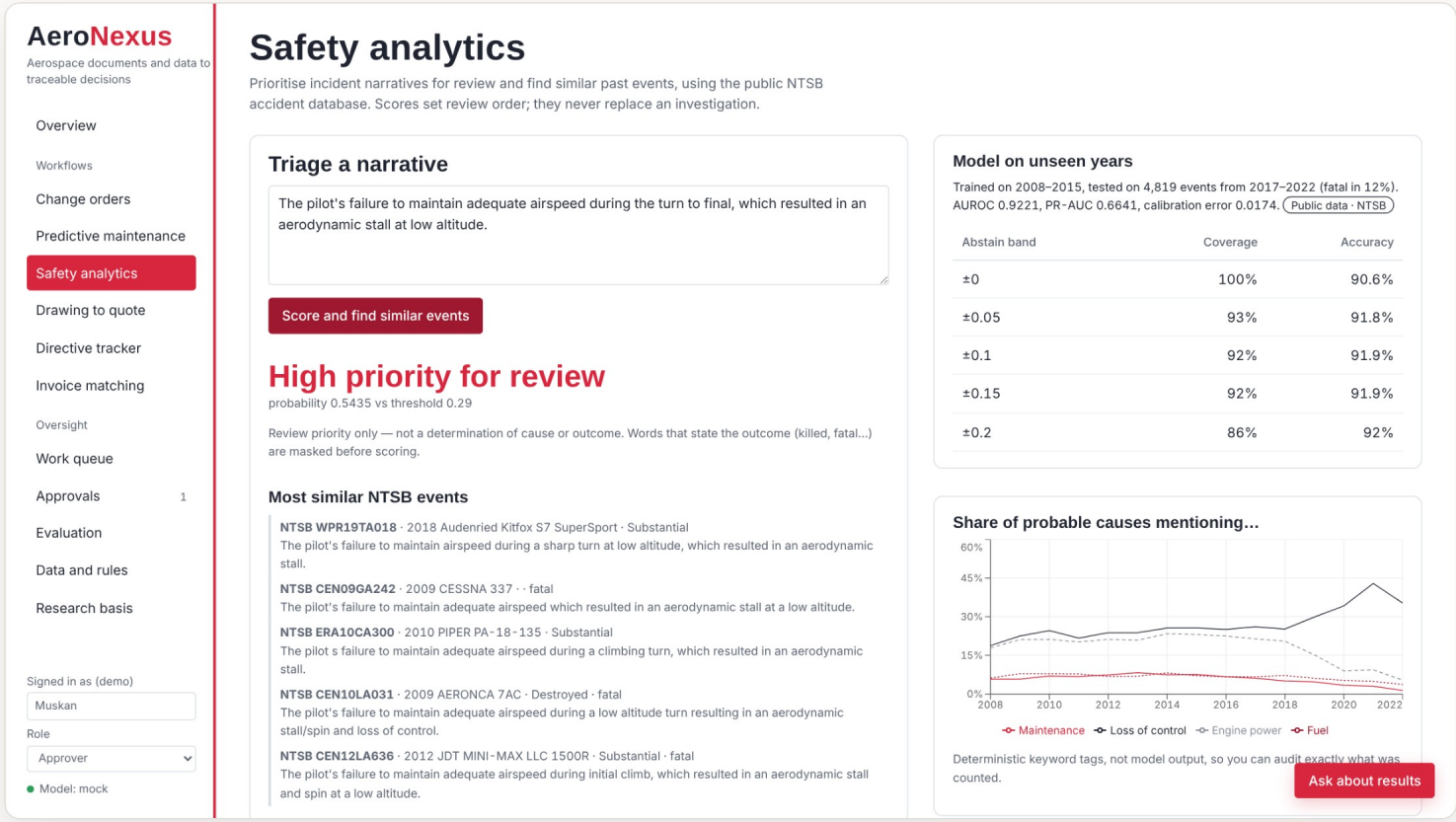
Train 2008–15, validate 2016, test on 2017–22.

Calibrated

AUROC 0.92, PR-AUC 0.66 at 12% prevalence, ECE 0.017.

Knows when to abstain

Uncertain cases go to a person, with similar past events.



WORKFLOW 4 · DRAWING TO QUOTE

Cheapest method first

Regex
Free and exact

→

NER
Gazetteer and model

→

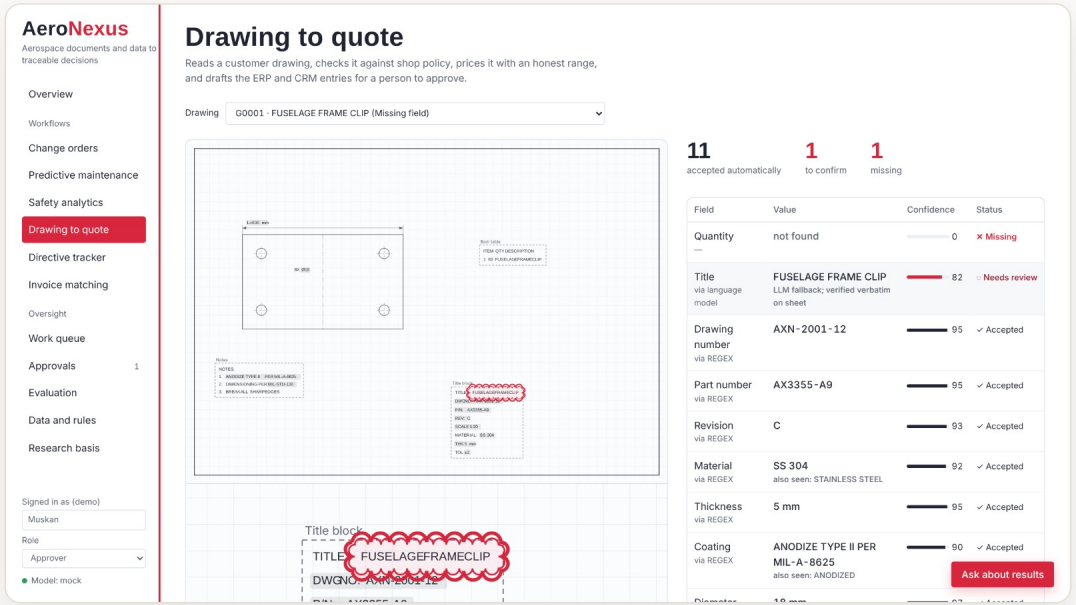
Llama 8B
One batched call

→

Llama 70B
Only when unsure

→

A person
Low confidence



Hallucination guard

An LLM value is kept only if it appears verbatim on the drawing.

Cross-checks

Title-block quantity against the BOM; a label with no value is flagged.

Then price it

Material and finish checks, a price with a conformal range, ERP and CRM drafts.

WORKFLOW 5 · INVOICE MATCHING

Match with the rule in force

Three-way match

Purchase order, goods receipt and invoice, line by line.

Effective-dated tolerance

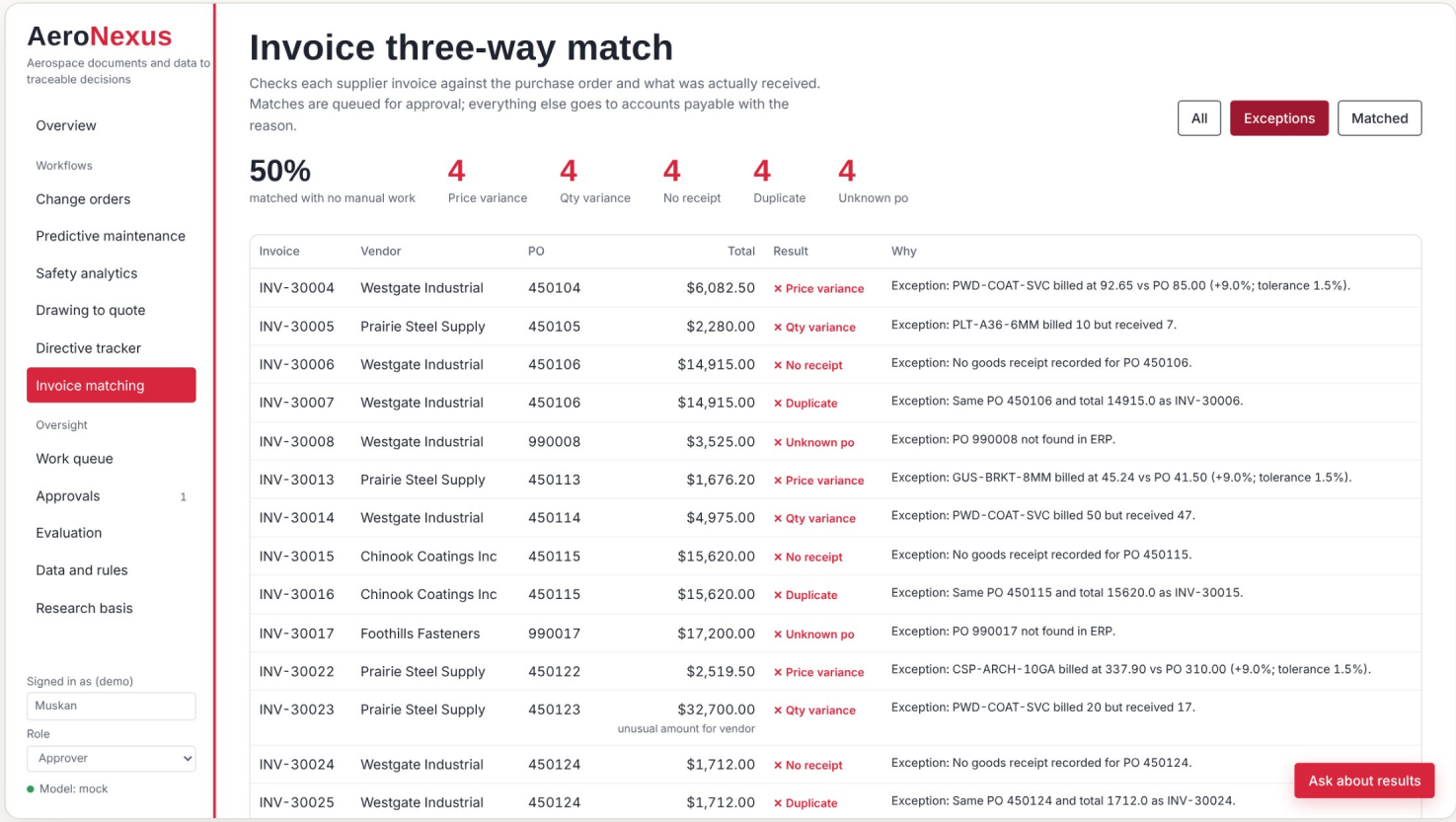
2% before 1 July 2026, 1.5% from then, by invoice date.

Explained exceptions

Each mismatch names the line, the rule and the variance.

50% touchless

Half the synthetic demo invoices pass with no human touch.



TRUST LAYER · GROUNDED ANSWERS

Answers that cite, or abstain

01 **Hybrid search:** BM25 plus embeddings, fused with RRF

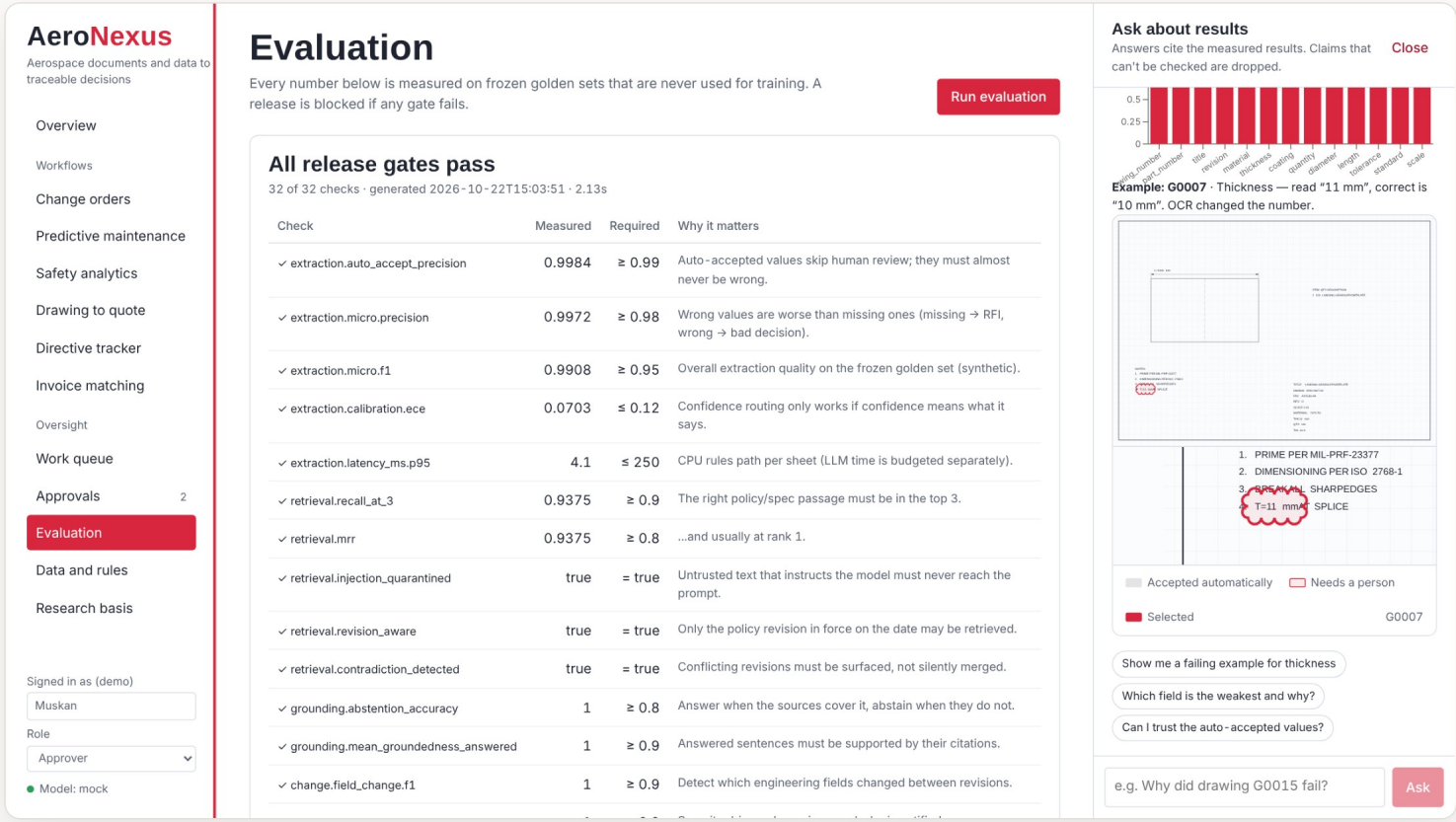
02 **Rerank and MMR** for varied evidence

03 **Revision-aware:** only the version in force on the date

04 **Quarantine** untrusted text; flag contradictions

05 **Check each sentence** against its citation

06 **Pass, prune or abstain** when evidence is missing



TRUST LAYER · EVALUATION

Any failing gate blocks a release

32 gates

Extraction, retrieval, grounding, change orders, models, security.

Frozen golden sets

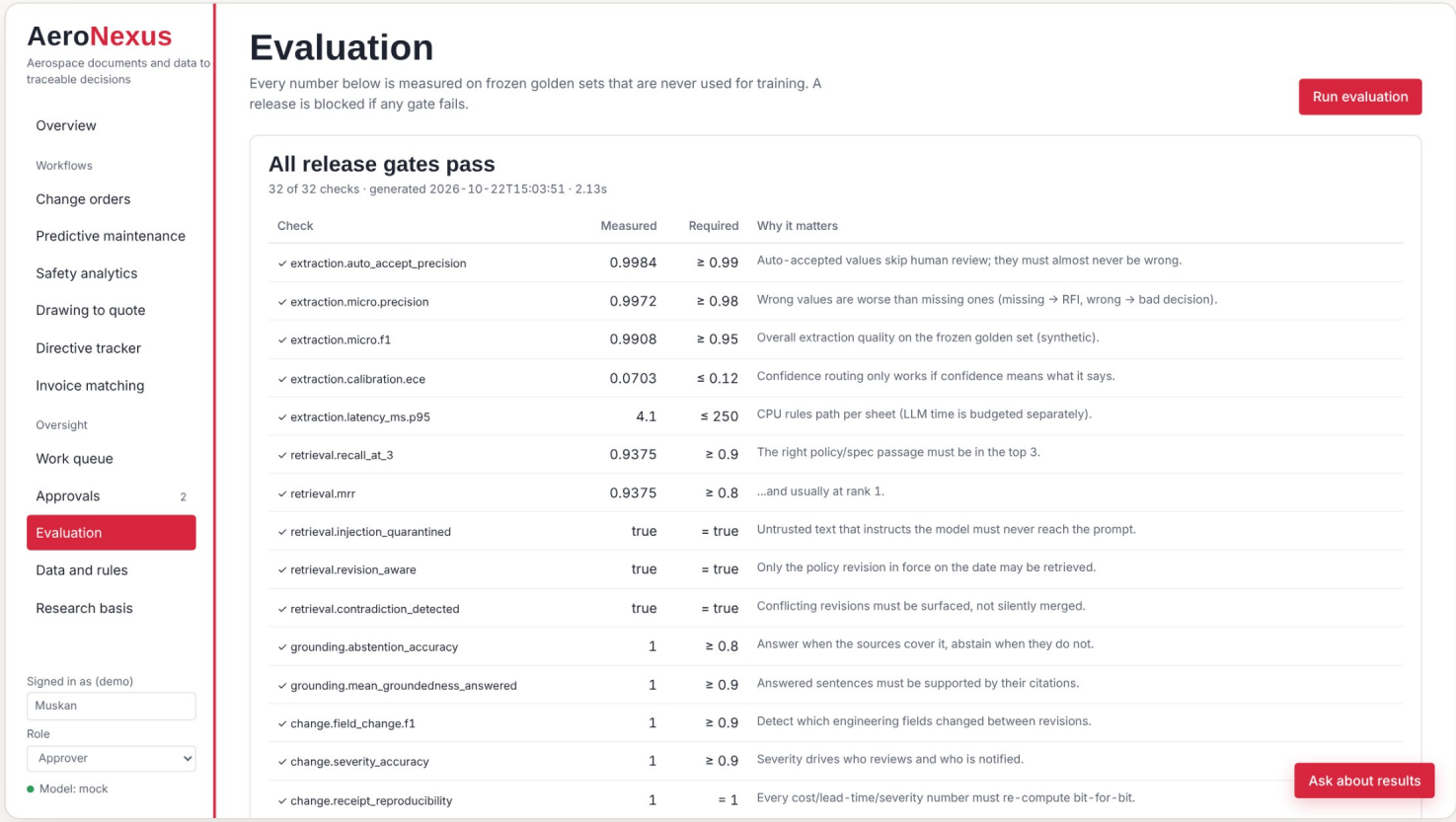
Never used for training; thresholds live in one file.

The judge is judged

Demo judge kappa 0.57 vs 0.68 between humans: refused.

Runs in CI

138 tests and the full gate on every change.



TRUST LAYER · RESULTS

Real data, kept apart from synthetic

Suite	Data	Result
Remaining life, FD001–FD004	Public · NASA C-MAPSS	RMSE 11.6 / 12.9 / 13.5 / 13.5 cycles; ranges cover 89 / 90 / 82 / 92%
Engine health score	Public · NASA C-MAPSS	AUROC 0.94–0.995
Safety triage	Public · NTSB 2017–22	AUROC 0.92 · PR-AUC 0.66 · ECE 0.017
Change orders	Synthetic	Field F1 1.0 · receipts and replays 100% identical
Drawing extraction	Synthetic	F1 0.99 · auto-accept precision 0.998
Access control	Tests	100% of role and tenant checks

RUNNING IT · FREE LLMS

Llama on Groq, used sparingly

Rules and NER first
Most work never reaches an LLM.

→

Small model
`llama-3.1-8b-instant`
Extraction fallback, answers, chat.

→

Large model
`llama-3.3-70b-versatile`
Only on low confidence or bad JSON.

Pacer
About 28 requests a minute, inside the free tier.

Rate limits
A 429 waits for retry-after; over 30 s, degrade.

Resilience
Cache, retries with backoff, circuit breaker, JSON repair.

Separate judge
gpt-oss-20b, a different family from the answers.

RUNNING IT · TRAINING

Train in Colab, test in VS Code

Colab
Notebooks 01–07 on TPU or GPU

→

Export
Models, reports and a manifest

→

Drive
One zip, then download

→

Import
Verify, back up, place, load

→

Gate
Tests and 32 checks decide

```
python scripts/import_models.py ~/Downloads/aeronexus_export.zip --verify
✓ SHA-256 match · version check · backup · smoke-load · pytest · eval gate
```

Or skip Colab

The baseline RUL and safety models train on a laptop CPU in about 4 minutes.

Pickle safety

Import only exports you made; a version mismatch prints the fix.

RUNNING IT · DATA

Free, open data, licences on record

Fetched and pinned

NASA C-MAPSS and NTSB, US Government works, SHA-256 pinned.

Manual exports

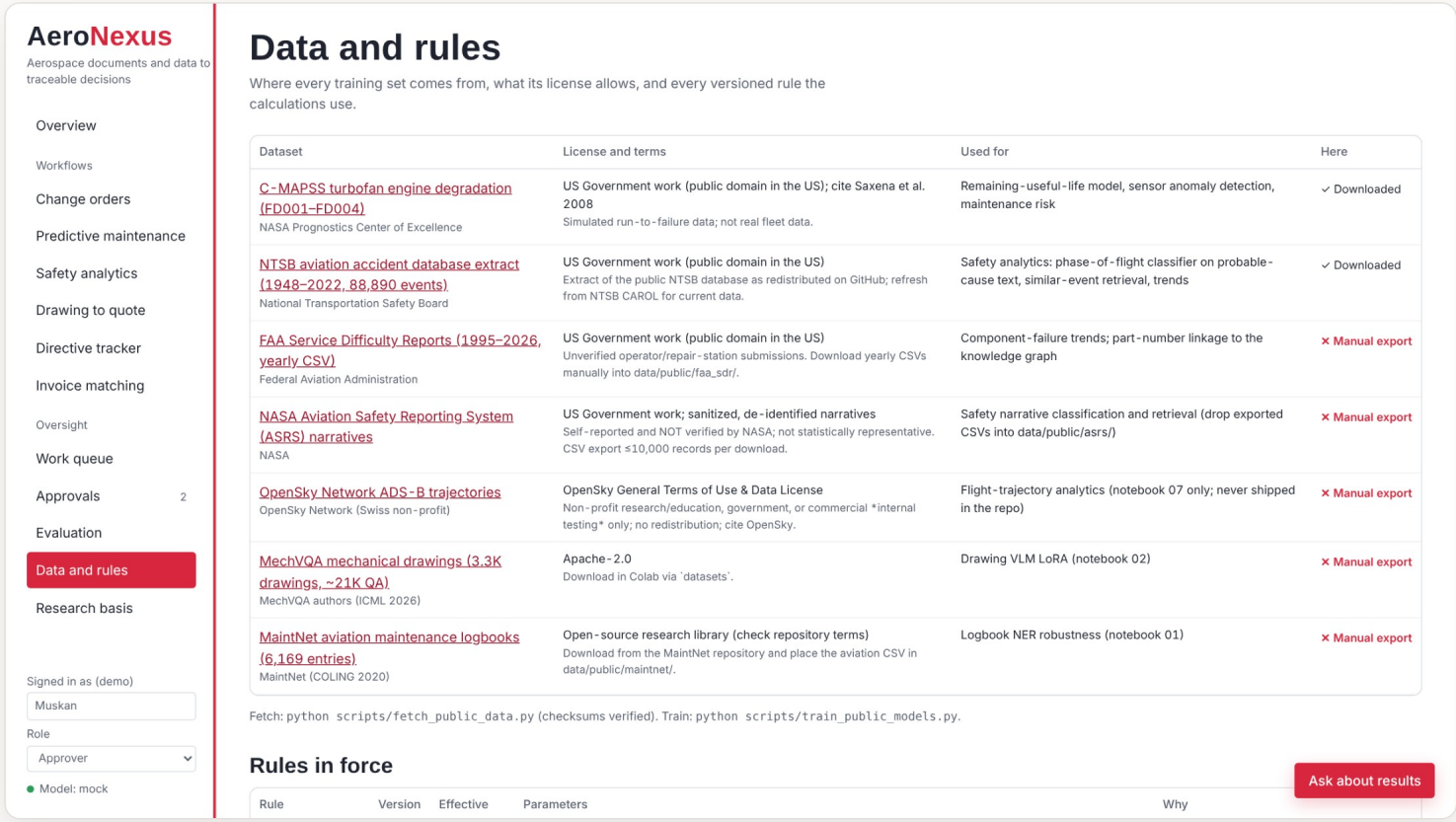
FAA SDR, NASA ASRS, MechVQA, MaintNet: terms recorded.

Not redistributed

OpenSky trajectories stay in Colab and are never shipped.

Synthetic, labelled

Drawings and ERP/CRM seed data are generated and marked.



HONEST LIMITS

Where it stands, and what comes next

Today

- Drawings, ERP and CRM data are synthetic
- C-MAPSS engines are simulated, not a real fleet
- Demo identity headers stand in for real sign-in
- The demo LLM judge is not trusted as a gate

Next

- Label real scanned drawings; re-calibrate gates
- OIDC sign-in; SAP and Salesforce adapters
- A/B the Colab deep RUL and transformer models
- Re-measure every gate with the Groq Llama models

AERONEXUS AI

Every number has a receipt. Every action has an approver.

```
docker compose up --build
```

Fetches the public data, trains, runs the 32 gates, then serves the app.

Thank you · Questions?